COSC 3326 – Homework 1

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1 Theoretical problems

Problem 1

a) The gradient of the image is given as

$$\nabla I(x,y) = \begin{bmatrix} \frac{\partial I}{\partial x} \\ \frac{\partial I}{\partial y} \end{bmatrix}$$

Therefore the image gradient magnitude is

$$||\nabla I(x,y)|| = \sqrt{(\frac{\partial I}{\partial x})^2 + (\frac{\partial I}{\partial y})^2}$$

b) Because the Sobel Operator approximates the partial derivatives of an image as G_x and G_y, the image gradient magnitude is

$$||\nabla I(x,y)|| = \sqrt{G_x^2 + G_y^2}$$

c) A higher threshold means that a larger gradient magnitude is needed to determine whether a pixel is an edge or not. With a higher threshold, less pixels would be considered as an edge, which would decrease the sensitivity and increase the specificity.

d) A high image gradient magnitude means the intensity for a given pixel quickly changes, this implies that a pixel with a high gradient magnitude is an edge.

e) The convolution kernels of the Sobel operator is given by:

$$S_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix} \qquad S_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

and the convolution kernels of the Prewitt operator is given by:

$$P_x = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \\ 1 & 0 & -1 \end{bmatrix} \qquad P_y = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ -1 & -1 & -1 \end{bmatrix}$$

Comparing the two kernels, the Sobel operator weights the pixels close to the center of the mask higher while Prewitt weights them equally.

Problem 2

a)

$$R = \det(M) - \alpha \operatorname{Trace}(M)^2$$

b)

$$\mathbf{M} = \sum_{(x,y) \in W} w(x,y) \begin{bmatrix} \frac{\partial I}{\partial x}^2 & \frac{\partial I}{\partial x}\frac{\partial I}{\partial y} \\ \frac{\partial I}{\partial y}\frac{\partial I}{\partial x} & \frac{\partial I}{\partial y}^2 \end{bmatrix}$$

c) The eigenvalues λ_1 and λ_2 of the structure tensor $\mathbf{M}$ describes how much the intensity changes if you shift the window in different directions. Two large eigenvalues determines that a given pixel is a corner.

d) With a higher threshold value, the corner response has to be greater in order for a pixel to be classified as a corner

e) The Harris Corner Detector response is invariant to rotation.